

Machine Learning-Based Classification of Medulloblastoma Using Gene Expression Data

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Abstract

Medulloblastoma is the most common malignant pediatric brain tumor, categorized into four molecular subgroups: WNT, SHH, Group 3, and Group 4. Current classification relies on broad genetic markers, often leading to misclassification and suboptimal treatment. This study applies basic machine learning techniques, including feature selection methods (Variance Thresholding, LASSO Regression, Random Forest Importance, and PCA) and classification models (SVM, Random Forest, XGBoost, and Logistic Regression) to predict subgroup classification based on gene expression data. Results indicate that feature selection enhances model performance, achieving high classification accuracy. The findings support the use of machine learning for improving medulloblastoma diagnosis and treatment planning.

1 Introduction

Medulloblastoma is a malignant childhood brain tumor, accounting for approximately 20% of all pediatric brain cancers [1]. It is divided into four subgroups (WNT, SHH, Group 3, and Group 4) based on molecular and genetic profiles [2]. Despite these distinctions, treatment remains largely uniform, leading to variable patient outcomes [3].

Machine learning offers a computational approach to enhance medulloblastoma classification by analyzing gene expression patterns. This study applies basic machine learning models to identify subgroup-specific genetic markers and improve classification accuracy.

2 Problem Statement

Current medulloblastoma treatment follows a generalized approach, despite known subgroup-specific genetic variations [4]. This can result in over-treatment for low-risk patients and under-treatment for high-risk cases, affecting survival rates [5].

Machine learning has shown promise in medical diagnosis, yet simple models have not been widely tested on medulloblastoma classification. The challenge lies in:

- Selecting the most informative genes while reducing high-dimensional data.
- Ensuring robust classification of subgroups with basic ML models.
- Evaluating the performance of different feature selection techniques.

This study focuses on addressing these challenges using traditional machine learning methods.

3 Dataset and Analysis

3.1 Dataset Overview

The dataset used in this study originates from [3], which provides comprehensive gene expression profiling of medulloblastoma tumors. This dataset consists of gene expression data from 73 medulloblastoma samples, each categorized into one of four established molecular subgroups: WNT, SHH, Group 3 (G3), and Group 4 (G4). These subgroups exhibit distinct molecular characteristics that influence their clinical behavior, prognosis, and treatment responses.

The original study leveraged next-generation sequencing to identify subgroup-specific mutations, transcriptional patterns, and epigenetic alterations, providing a foundation for subgroup classification based on gene expression. By utilizing this dataset, we aim to develop machine learning models that can accurately classify medulloblastoma subgroups based on these molecular profiles.

The dataset comprises:

- **54,675 features (genes)** representing normalized gene expression levels across all samples.
- **73 samples (patients)**, categorized into the following subgroups:
 - 16 samples labeled as G3.
 - 29 samples labeled as G4.
 - 10 samples labeled as SHH.
 - 8 samples labeled as WNT.

3.2 Preprocessing and Feature Engineering

Prior to applying machine learning models, we performed the following preprocessing steps:

- **Normalization:** Gene expression values were processed to ensure comparability across samples.
- **Handling Missing Values:** Any missing or anomalous values were addressed to maintain data integrity.
- **Label Extraction:** Sample names were parsed to extract subgroup labels (e.g., 'G3_1' → 'G3').
- **Feature Selection and Dimensionality Reduction:** To mitigate the curse of dimensionality, we applied multiple feature selection techniques:
 - **Variance Thresholding:** Removing genes with low variance across samples.
 - **LASSO-based Feature Selection:** Selecting genes with strong predictive value using L1-regularized regression.
 - **Tree-Based Feature Selection:** Utilizing Random Forest to rank genes by importance.
 - **Principal Component Analysis (PCA):** Transforming gene expression profiles into lower-dimensional representations.

3.3 Statistical Insights

To assess the complexity and variance within the dataset, we examined:

- The distribution of gene expression values across subgroups.
- The number of features retained after each feature selection method.
- The impact of dimensionality reduction on classification performance.

These analyses ensure that only the most relevant features are utilized, optimizing model performance while reducing computational overhead.

4 Methodology

4.1 Machine Learning Models

Four classification models were trained and evaluated:

- **Support Vector Machine (SVM)** – Linear kernel, handles high-dimensional data well.
- **Random Forest (RF)** – Ensemble method to improve generalization.
- **XGBoost** – Gradient boosting for robust classification.
- **Logistic Regression (LR)** – A strong baseline model.

4.2 Experimental Setup

- **Dataset split:** 80% training, 20% testing (stratified).
- **Evaluation metrics:** Accuracy, F1-score for each class.
- **Feature selection applied before training** to reduce overfitting.

5 Results

The models were evaluated across 22 different datasets, each corresponding to a different feature selection method. The evaluation focused on accuracy and F1-scores for each medulloblastoma subgroup (G3, G4, SHH, WNT).

5.1 Overall Best Model Performance

To determine which models performed best across different datasets, we counted the number of times each model achieved the highest accuracy, as shown in 1, considering cases where multiple models had the same best performance.

From the analysis, Logistic Regression and SVM emerged as the most frequently best-performing models, achieving the highest accuracy in 21 and 20 datasets, respectively. Random Forest and XGBoost followed with 13 and 8 wins, respectively. This suggests that simpler linear models like Logistic Regression and SVM generalized well across different feature selection methods, while tree-based methods like Random Forest and XGBoost exhibited more variability.

	SVM	RandomForest	XGBoost	LogisticRegression	Total Datasets
Best Model Count	20	13	8	21	22

Table 1: Best Model Performance Count Across Datasets (Considering Ties).

5.2 Performance Across Feature Selection Methods

We further analyzed model performance across different feature selection methods: LASSO (L1) Regression, Random Forest Feature Importance, Variance Thresholding, and PCA.

5.2.1 LASSO (L1) Regression-Based Feature Selection

LASSO regression-based feature selection reduced the dataset to a smaller subset of the most predictive genes while maintaining high classification performance. Based on the results obtained, as shown in 2, Logistic Regression and SVM performed consistently well across different LASSO thresholds, achieving accuracy levels above 93% in most cases. XGBoost, however, showed a decline in performance when fewer features were selected, likely due to the model’s reliance on a richer feature space for effective learning. Random Forest exhibited more variance in its accuracy, suggesting that its feature importance ranking may not always align optimally with the selected LASSO features.

α	n_feature	Model	Acc.	F1 G3	F1 G4	F1 SHH	F1 WNT
1e-2	66	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.8667	0.75	1.0	1.0	0.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
1e-3	75	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.8667	0.6667	0.9412	1.0	0.6667
		XGBoost	0.7333	0.5	0.875	1.0	0.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
1e-4	85	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.9333	0.8	0.9412	1.0	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
1e-5	204	SVM	1.0	1.0	1.0	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.8667	0.6667	0.9412	1.0	0.6667
		LogisticRegression	1.0	1.0	1.0	1.0	1.0
1e-6	649	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.8667	0.8	0.8889	0.6667	1.0
		XGBoost	0.8667	0.6667	0.9412	0.6667	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0

Table 2: Performance of ML models using LASSO (L1) Feature Selection.

5.2.2 Random Forest Feature Selection

Random Forest feature selection ranked features based on importance and selected the most relevant genes for classification. Results as shown in 3 showed that increasing the number of trees ('n_estimators') led to more stable feature selection, with better generalization in SVM and Logistic Regression. While Random Forest performed well in most cases, its reliance on tree splits sometimes led to inconsistencies in classification, particularly when dealing with highly correlated features. XGBoost exhibited mixed performance, benefitting in some cases but struggling in others when the feature space was reduced too aggressively.

n_estim	n_feature	Model	Acc.	F1 G3	F1 G4	F1 SHH	F1 WNT
25	288	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.8667	0.5	0.8889	1.0	1.0
		XGBoost	0.8	0.8	0.8421	0.6667	0.6667
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
50	559	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.8	0.8	0.8421	1.0	0.0
		XGBoost	0.8667	0.8571	0.9412	0.6667	0.6667
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
100	1145	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.8667	0.8	0.8889	0.6667	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
200	2307	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	1.0	1.0	1.0	1.0	1.0
		XGBoost	0.9333	1.0	0.9412	1.0	0.6667
		LogisticRegression	1.0	1.0	1.0	1.0	1.0
300	3371	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.8667	0.8	0.9412	0.5	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0

Table 3: Performance of ML models using Random Forest Feature Selection.

5.2.3 Variance Threshold Feature Selection

Variance thresholding removed genes with minimal expression variability across samples. Results as shown in 4 showed that reducing the dataset by eliminating low-variance features had little negative impact on model performance. SVM and Logistic Regression remained highly accurate across different thresholds, while XGBoost and Random Forest saw moderate fluctuations. Notably, when the variance threshold was set too high (e.g., retaining only 75 features), Random Forest and XGBoost struggled due to the loss of critical information, whereas SVM and Logistic Regression remained robust.

Threshold	n_feature	Model	Acc.	F1 G3	F1 G4	F1 SHH	F1 WNT
0.03	54531	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.8667	0.8	0.8889	0.6667	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
0.1	50160	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.8667	0.8	0.8889	0.6667	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
0.3	17394	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.8667	0.8	0.8889	0.6667	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
1.0	1923	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.9333	1.0	0.9412	0.6667	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
3.0	75	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	1.0	1.0	1.0	1.0	1.0
		XGBoost	0.8667	0.75	1.0	1.0	0.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0

Table 4: Performance of ML models using Variance Threshold Feature Selection.

5.2.4 Principal Component Analysis (PCA)

PCA-based feature selection reduced dimensionality while preserving variance. Results as shown in 5 indicate that SVM and Logistic Regression maintained consistently high performance across different numbers of principal components, with accuracy around 93%. Random Forest and XGBoost, however, suffered when too few components were retained, as decision trees and boosting methods rely on structured feature spaces rather than transformed components. When at least 20 principal components were retained, all models performed comparably, suggesting that this threshold effectively captured most of the relevant variation in the data.

n_compon	n_feature	Model	Acc.	F1 G3	F1 G4	F1 SHH	F1 WNT
1	1	SVM	0.8667	0.8	0.9412	0.8	0.6667
		RandomForest	0.8	0.6667	0.9412	0.5	0.6667
		XGBoost	0.7333	0.6667	0.9412	0.4	0.0
		LogisticRegression	0.8667	0.8	0.9412	0.8	0.6667
5	5	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.9333	0.8	0.9412	1.0	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
10	10	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.9333	0.8	0.9412	1.0	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
20	20	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.8667	0.8	0.8889	1.0	0.6667
		XGBoost	0.9333	0.8	0.9412	1.0	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
30	30	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.9333	0.8	0.9412	1.0	1.0
		XGBoost	0.9333	0.8	0.9412	1.0	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
50	50	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.8667	0.8	0.8889	1.0	0.6667
		XGBoost	0.9333	0.8	0.9412	1.0	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0
73	73	SVM	0.9333	0.8	0.9412	1.0	1.0
		RandomForest	0.8	0.8	0.8421	0.6667	0.6667
		XGBoost	0.9333	0.8	0.9412	1.0	1.0
		LogisticRegression	0.9333	0.8	0.9412	1.0	1.0

Table 5: Performance of ML models using PCA for Dimensionality Reduction.

5.3 Key Findings

- Logistic Regression and SVM were the most reliable models, achieving high accuracy across all feature selection methods.
- Random Forest performed well when enough features were retained, but struggled in datasets with aggressive feature selection.
- XGBoost exhibited the most variance in performance, benefiting in some cases but underperforming when features were too restricted.
- Feature selection methods did not significantly impact SVM and Logistic Regression, while tree-based models were more sensitive to reduced feature spaces.
- Variance Thresholding and PCA were the most stable methods, retaining key information while improving efficiency.

5.4 Implications and Recommendations

These results suggest that Logistic Regression and SVM should be preferred for medulloblastoma classification when computational efficiency and robustness are priorities. Random Forest and XGBoost may still be viable choices, particularly when dealing with larger feature sets where their ability to capture non-linear relationships becomes beneficial.

For future studies, combining multiple feature selection techniques (e.g., using LASSO followed by PCA) could provide better feature representations, potentially improving classification performance across all models. Additionally, exploring ensemble methods that integrate SVM and tree-based models could yield a more balanced approach to medulloblastoma classification.

6 Conclusion

This study demonstrates that machine learning models can effectively classify medulloblastoma subgroups based on gene expression data. By leveraging feature selection techniques, we significantly improved classification performance, reducing dimensionality while retaining the most informative genes. Our results indicate that SVM and Logistic Regression consistently achieved high classification accuracy, while Random Forest and XGBoost showed more variability depending on feature selection parameters.

Key Takeaways:

- SVM and Logistic Regression provided the most consistent and robust performance across different feature selection methods.
- Random Forest achieved strong results but was highly sensitive to the number of selected features.
- XGBoost struggled with class imbalance, suggesting the need for further hyperparameter tuning or oversampling techniques.

While our study provides an effective pipeline for molecular subgroup classification, it is limited by the availability of positive samples (tumor-containing datasets) and the relatively small dataset size.

Future work should focus on:

- Incorporating negative control datasets (samples without brain tumors) to better distinguish between medulloblastoma subgroups and non-tumor cases.
- Exploring deep learning architectures, such as convolutional and recurrent neural networks, to capture complex gene expression patterns.
- Expanding the dataset with additional sequencing studies to enhance generalizability and model robustness.
- Further optimizing feature selection thresholds to ensure that selected genes contribute meaningfully to classification without overfitting.

By addressing these limitations, future research can further refine machine learning-based classification models, aiding in early diagnosis and personalized treatment strategies for medulloblastoma patients.

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